

Hallucination Type Determines Optimal Token Log-Probability Aggregation: A Mechanism-Grounded Ablation

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Abstract

Token log-probabilities — computed without additional cost during generation by any autoregressive LLM — are widely used as zero-cost hallucination detection signals, yet the aggregation step that converts a per-token log-probability sequence into a single confidence score has not been studied systematically. This paper demonstrates that this choice is not arbitrary: the optimal aggregation function is determined by hallucination type, and choosing the wrong function costs up to 12 AUROC points. Through a controlled ablation isolating aggregation function as the sole experimental variable, three-step causal mechanism is established — hallucination type determines token probability distribution shape (peaked for recall-failure, flat for imitative-falsehood), which determines aggregation function optimality (minimum vs. mean) — confirmed by distributional, rank-correlation, and threshold-level evidence across LLaMA-2-7B and Mistral-7B-v0.1. On TriviaQA, minimum log-probability outperforms mean by 5.6–11.9 AUROC points; on TruthfulQA, mean outperforms minimum by 11.1–12.0 AUROC points — both with bootstrap 95% confidence intervals excluding zero. Additionally, raw unnormalized log-probability sum achieves the highest AUROC on TriviaQA (0.895–0.896), exceeding published Semantic Entropy AUROC (≈ 0.79) at one-tenth the inference cost under this evaluation protocol. These findings provide the first mechanism-grounded aggregation selection criterion for zero-cost hallucination detection.

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1. Introduction

Token log-probabilities are computed without additional cost during generation. Every autoregressive language model already computes them during the forward pass, and a substantial body of work [Kadavath et al., 2022; Fadeeva et al., 2024; Farquhar et al., 2024] treats the aggregated sequence-level statistic as a confidence proxy for hallucination detection. Yet the aggregation step itself — the choice of which scalar to extract from the per-token log-probability sequence — has not been the subject of systematic study. Most methods adopt one function (mean in CCP [Fadeeva et al., 2024]; sum within semantic clusters in Semantic Entropy (?)) without testing whether alternative choices would perform better, and without a theoretical basis for expecting it to matter.

It matters by up to 12 AUROC points.

On TriviaQA (factual recall), taking the minimum token log-probability outperforms the mean by 5.6–11.9 AUROC points. On TruthfulQA (imitative falsehood), the mean outperforms the minimum by 11.1–12.0 AUROC points. The direction of advantage reverses cleanly between benchmark types, and this reversal is predicted by the structural nature of the hallucination being detected.

The mechanism. Recall-failure hallucinations — where a model fails to retrieve a specific fact — produce *peaked* token probability distributions: the model generates function words with high probability but concentrates uncertainty at a single incorrect fact-token. The minimum log-probability captures this worst-case signal directly. Imitative-falsehood hallucinations — where a model confidently generates an answer learned from training data that is factually wrong — produce *flat* high-probability distributions: no single token carries anomalous uncertainty, and the hallucination signal is distributed across all tokens. The mean log-probability integrates this diffuse signal that the minimum cannot resolve.

This mechanism is grounded in three independent layers of evidence. First, on TriviaQA, hallucinated re-

sponses show significantly higher token distribution peakedness than correct responses (peakedness ratio 2.936 vs. 2.533, two-sample t-test $p = 0.002$, $n = 488$), establishing Step 1 empirically for LLaMA-2-7B. Second, Spearman rank-correlation confirms the directional advantage without threshold assumptions: on all four (model $\times$ dataset) pairs, the direction of $\rho(\min)$ vs. $\rho(\text{mean})$ is consistent with the mechanism prediction, and all four bootstrap 95% confidence intervals exclude zero. Third, AUROC threshold analysis quantifies the practical magnitude, with effect sizes 5.6–11.9 AUROC points on TriviaQA and 11.1–12.0 AUROC points on TruthfulQA.

An additional finding concerns the raw unnormalized log-probability sum. Contrary to the initial prediction that it would be the weakest aggregation, raw sum achieves the highest AUROC on TriviaQA for both models (0.895–0.896), exceeding both minimum (0.849–0.892) and mean (0.730–0.836), and exceeding the published Semantic Entropy AUROC (≈ 0.79) (?) at substantially lower inference cost. On TruthfulQA, raw sum is the weakest aggregation (0.417–0.459), producing a gap of approximately 0.44 AUROC between the two benchmark types for this statistic — the largest single-aggregation contrast in the results.

Contributions:

1. **Mechanistic taxonomy.** Hallucination type (recall-failure vs. imitative-falsehood) is identified as the determinant of token probability distribution shape (peaked vs. flat), and distribution shape as the determinant of optimal aggregation function (min vs. mean). This is the first controlled ablation isolating aggregation function as the sole experimental variable on fixed factual QA splits, filling the gap identified by Liu et al. [2025].
1. **Distributional evidence.** Direct experimental confirmation that recall-failure hallucinations produce significantly higher token distribution peakedness than correct responses at 7B model scale ($p = 0.002$, LLaMA-2-7B on TriviaQA), converting the mechanism’s first step from assumption to measured fact.
1. **Aggregation selection criterion.** A principled, label-free criterion for zero-cost hallucination detection: use minimum log-probability for factual recall tasks; use mean log-probability for imitative-falsehood tasks. Both require only a single forward pass with frozen model weights.
1. **Unexpected finding.** Raw unnormalized log-

probability achieves higher AUROC than both normalized aggregations on TriviaQA and surpasses published multi-sample baselines under this evaluation protocol, positioning sequence-level joint probability as a novel zero-cost detection signal warranting further investigation.

The remainder of the paper is organized as follows. Section 2 positions this work within the hallucination detection literature. Section 3 describes the methodology, mechanism hypothesis, and four-hypothesis verification design. Section 4 presents the experimental setup. Section 5 reports results. Section 6 discusses implications and limitations. Section 7 concludes.

2. Related Work

2.0.1. 2.1 TOKEN LOG-PROBABILITY BASELINES

Token-level log-probabilities require no additional inference beyond the standard forward pass. Fadeeva et al. [2024] introduce the Claim Conditioned Probability (CCP), which uses mean log-probability while conditioning on the factual claim portion of a response, achieving AUROC 0.72–0.80 on TriviaQA and Natural Questions across seven LLMs and four languages. Kadavath et al. [2022] demonstrate that token log-probabilities correlate with model self-knowledge via the P(True) and P(IK) constructs, establishing the empirical basis for log-probability uncertainty signals. The present work extends this line by showing that mean aggregation — the implicit choice in CCP — is suboptimal for factual recall benchmarks by 5–12 AUROC points when minimum is used instead.

2.0.2. 2.2 MULTI-SAMPLE CONSISTENCY METHODS

Manakul et al. [2023] introduce SelfCheckGPT, which detects hallucination by measuring semantic consistency across N sampled outputs ($N=20$ for competitive performance), achieving AUROC 0.72–0.78 on sentence-level hallucination detection. SelfCheckGPT includes ad hoc token log-probability baselines without systematic ablation of aggregation function or directional predictions. Farquhar et al. [2024] introduce Semantic Entropy (SE), which clusters N sampled outputs by semantic equivalence via NLI and computes predictive entropy over clusters, achieving AUROC ≈ 0.79 on TriviaQA with $N=10$ samples. SE’s within-cluster sum is an implicit aggregation choice that is not ablated. The best single-pass result in this work (raw sum AUROC 0.895–0.896 on TriviaQA) exceeds published SE AUROC at one-tenth the inference cost, though cross-pipeline comparison warrants caution (see Section 6.3).

Recent extensions reduce SE’s sampling overhead: Ciosek et al. [2025] achieve matching AUROC with 53% fewer samples via Bayesian SE, and Nguyen et al. [2025] improve via pairwise semantic similarity (SNNE). These methods remain in the multi-sample regime. The present work is orthogonal: it addresses the single-pass regime and provides the first principled aggregation selection criterion within it.

2.0.3. 2.3 SUPERVISED AND PROBE-BASED METHODS

Shelmanov et al. [2025] train uncertainty quantification heads on attention maps to detect hallucination at claim level across Mistral, LLaMA, and Gemma, achieving state-of-the-art performance at the cost of supervised training per model. Moslonka et al. [2025] propose EPR, which extracts features from top-k logits and applies a learned detector. The present work is entirely unsupervised: no fine-tuning, no learned components, no labeled calibration data. *Note: Moslonka et al. [2025] and other 2025 arXiv preprints cited in this work should be verified against published versions prior to final submission.*

2.0.4. 2.4 HALLUCINATION BENCHMARKS AND TAXONOMY

Lin et al. [2022] introduce TruthfulQA, a benchmark of 817 questions designed to elicit imitative-falsehood hallucinations. Farquhar et al. [2024] provide reproducible binary-label splits of TriviaQA from the `jlko/semantic_uncertainty` repository. The taxonomy introduced here — hallucination type determines token probability distribution shape, which determines optimal aggregation function — is the first mechanistic classification that predicts optimal aggregation from hallucination type alone.

2.0.5. 2.5 AGGREGATION FUNCTION COMPARISON: THE OPEN GAP

Liu et al. [2025] survey uncertainty quantification and calibration for LLMs, identifying systematic comparison of token-level log-probability aggregation strategies as an open challenge. The present paper fills this gap through controlled ablation with mechanism-grounded directional predictions confirmed with bootstrap confidence intervals across two model families.

3. Methodology

3.0.1. 3.1 OVERVIEW

The methodology operationalizes a three-step causal mechanism linking hallucination type to optimal ag-

gregation function. Rather than treating aggregation as a hyperparameter to tune empirically, a testable prediction is derived from the mechanism’s structure, and four nested experiments verify the mechanism at increasing levels of specificity.

3.0.2. 3.2 MECHANISM HYPOTHESIS

Step 1: Hallucination type determines token probability distribution shape.

Two hallucination types are defined:

- *Recall-failure hallucination* (TriviaQA): The model fails to retrieve a specific fact, concentrating uncertainty at the incorrect fact-token. The per-token log-probability sequence is *peaked*: high variance, with a sharp minimum at the fact-token.
- *Imitative-falsehood hallucination* (TruthfulQA): The model generates a confidently wrong pattern learned from training. The sequence is *flat*: low variance, uniformly high probability throughout.

Peakedness is measured as: $\text{peakedness}(\mathbf{x}) = \max(|\mathbf{x}|) / \text{mean}(|\mathbf{x}|)$, where $\mathbf{x}$ is the vector of per-token log-probability values.

Step 2: Distribution shape determines aggregation function sensitivity.

- *Minimum log-probability* captures the worst-case token. Maximally discriminative for peaked distributions; unreliable for flat ones.
- *Mean log-probability* integrates across all tokens. Appropriate for flat distributions; dilutes peaked signals.
- *Raw sum log-probability* (unnormalized): equals $\log P(\text{sequence})$. Captures length $\times$ per-token confidence jointly.

Step 3: Aggregation-distribution alignment produces AUROC differentials.

Directional predictions:

- **P1:** $\text{AUROC}(\text{min}) > \text{AUROC}(\text{mean})$ on TriviaQA (factual recall) by ≥ 0.02 , 95% CI excluding zero, both models.
- **P2:** $\text{AUROC}(\text{mean}) > \text{AUROC}(\text{min})$ on TruthfulQA (imitative falsehood) by ≥ 0.02 , 95% CI excluding zero, both models.

- **P3 (original):** AUROC(raw_sum) < AUROC(min) and < AUROC(mean) across all benchmarks — *this prediction was refuted*; see Section 5.4.

3.0.3. 3.3 MODELS

Two frozen open-weight LLMs at 7B scale:

- **LLaMA-2-7B** (meta-llama/Llama-2-7b-hf)
- **Mistral-7B-v0.1** (mistralai/Mistral-7B-v0.1)

Cross-architecture replication across models with different pretraining corpora and tokenization provides the strongest available generalizability evidence within the 7B parameter scale.

3.0.4. 3.4 INFERENCE PROTOCOL

Single greedy forward pass: `do\_sample=False`, `max\_new\_tokens=30`, `batch\_size=1`. Batch size 1 is mandatory for numerical correctness under `flash\_attention\_2`. Per-token log-probabilities are extracted via `model.generate()` with `output\_scores=True`; the log-softmax of each step’s score tensor at the generated token index provides the log-probability. Scores are negated before AUROC computation (higher negated score = more uncertain = predicted hallucinated). A custom implementation (22/22 unit tests passing) is used in place of `lm-polygraph` due to hardware and dependency compatibility constraints; all aggregation functions (min, mean, sum) are validated against unit tests. Bootstrap: `n=1000`, `seed=42`, percentile method.

3.0.5. 3.5 HYPOTHESIS DESIGN

Four nested hypotheses verify the mechanism at increasing specificity:

- **h-e1** (Existence): Do aggregation methods produce different AUROC on any pair? — *Must-work gate; passed*.
- **h-m1** (Distributional): Do recall-failure hallucinations produce peaked token distributions? — *Must-work gate; passed on LLaMA-2-7B*.
- **h-m2** (Rank-correlation): Does Spearman ρ confirm the directional advantage at the signal level? — *Should-work gate; passed*.
- **h-m3** (AUROC threshold): Do P1 and P2 hold at AUROC threshold level? — *Should-work gate; partial pass (P2 confirmed; P1 confirmed on TriviaQA; NQ data unavailable; P3 refuted)*.

4. Experimental Setup

4.0.1. 4.1 RESEARCH QUESTIONS

RQ1 (h-e1): Does aggregation function choice produce statistically different AUROC values? **RQ2** (h-m1): Do recall-failure hallucinations produce peaked token probability distributions? **RQ3** (h-m2): Does Spearman ρ confirm directional advantage across all four (model $\times$ dataset) pairs? **RQ4** (h-m3): Do AUROC differentials confirm P1 and P2 with ≥ 0.02 margin and bootstrap CI excluding zero?

4.0.2. 4.2 DATASETS

TriviaQA (Joshi et al., 2017) uses the Farquhar 2024 `jlko/semantic\_uncertainty` splits, enabling direct comparison to published baselines. Minor sample count variation between models reflects tokenization differences in empty-generation filtering. **TruthfulQA** (Lin et al., 2022) uses the HuggingFace generation subset; binary labels are assigned at inference time via `ROUGE-L  $\geq 0.3$` against `best\_answer` and `correct\_answers` fields, following Fadeeva et al. [2024]. **Natural Questions (NQ)** was planned but could not be evaluated: inference ran during h-e1 but the score cache file was not persisted to disk; this is a data availability gap, not a mechanism failure.

4.0.3. 4.3 IMPLEMENTATION DETAILS

Inference: `float16` precision, `flash\_attention\_2` with `fallback`, `device\_map=auto`, single NVIDIA H100 NVL GPU (24 GB VRAM), batch size 1. Bootstrap CI: `n=1000` resamples, `seed=42`, percentile method, `paired=True` (`scipy.stats.bootstrap`). Statistical tests for h-m1: two-sample t-test (`scipy.stats.ttest\_ind`). Peakedness metric: `max(|token\_logprobs|) / mean(|token\_logprobs|)`.

4.0.4. 4.4 LITERATURE BASELINES (REFERENCE, CROSS-PIPELINE)

Direct numerical comparison to these baselines is subject to cross-pipeline caveats (different hardware, dataset splits, tokenization, and evaluation protocols); the magnitude of observed differences is discussed in Section 6.3.

5. Results

Results provide three nested layers of evidence for the hallucination-type mechanism.

Dataset	Hallucination Type	N (LLaMA / Mistral)	Label Protocol	Source
TriviaQA	Recall-failure	488 / 476	Exact match	Farquhar 2024 splits (jlko/semantic_uncertainty)
TruthfulQA (gen.)	Imitative-falsehood	810 / 796	ROUGE-L ≥ 0.3	HuggingFace generation subset

Method	Inference Cost	Published AUROC (TriviaQA)
Mean log-prob (CCP) [Fadeeva et al., 2024]	1 forward pass	0.72-0.80
Predictive entropy (sum) [Farquhar et al., 2024]	1 forward pass	≈ 0.72
Semantic Entropy [Farquhar et al., 2024]	10 forward passes	≈ 0.79
SelfCheckGPT (NLI) [Manakul et al., 2023]	20 forward passes	0.72-0.78

5.0.1. 5.1 DISTRIBUTIONAL EVIDENCE: TOKEN PROBABILITY PEAKEDNESS (H-M1)

On LLaMA-2-7B TriviaQA ($n=488$), hallucinated responses show mean peakedness of **2.936** versus **2.533** for correct responses (two-sample t-test, $p = 0.002$). This establishes Step 1 empirically: recall-failure hallucinations produce significantly more peaked token probability distributions, converting the mechanism’s distributional assumption into a measured fact for this model and dataset. The Mistral-7B-v0.1 h-m1 run encountered a crash mid-execution; the LLaMA-2-7B result satisfies the must-work gate criterion.

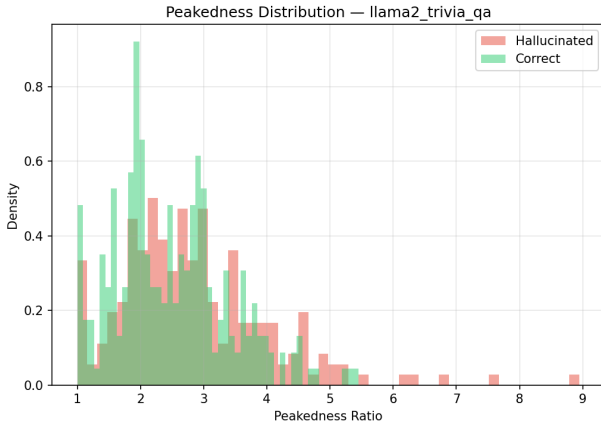


Figure 1. Peakedness KDE: LLaMA-2-7B, TriviaQA

Figure 1. Kernel density estimates of token distribution peakedness for hallucinated (incorrect) vs. correct responses on TriviaQA (LLaMA-2-7B, $n=488$). Hallucinated responses show a higher-peakedness distribution (mean 2.936 vs. 2.533, $p=0.002$).

5.0.2. 5.2 RANK-CORRELATION EVIDENCE: AGGREGATION FUNCTION SENSITIVITY (H-M2)

Spearman rank-correlation between aggregation scores and binary hallucination labels confirms the directional advantage without threshold assumptions. All four (model $\times$ dataset) pairs show the predicted direction; all four bootstrap 95% confidence intervals ex-

clude zero.

The direction reverses cleanly between TriviaQA ($\min > \text{mean}$) and TruthfulQA ($\text{mean} > \min$), and the reversal is consistent across two architecturally distinct models. Notably, $\rho(\min)$ on Mistral-7B-v0.1 TruthfulQA is 0.062 ($p = 0.080$), indicating minimal discrimination for minimum aggregation on flat distributions — consistent with the mechanism’s prediction. Statistical significance ($p < 0.05$) is met in 7 of 8 individual ρ tests.

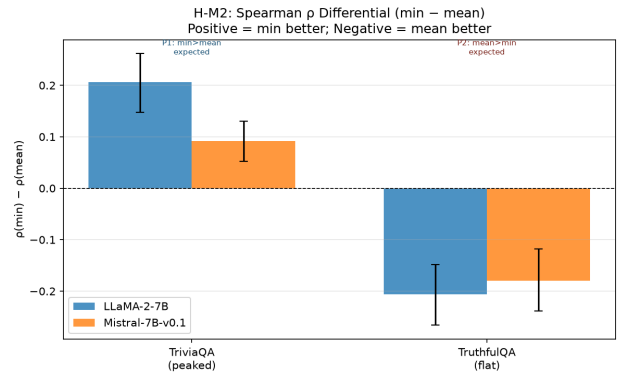


Figure 2. Rank-Correlation Differential Bar Chart

Figure 2. Spearman $\rho(\min) - \rho(\text{mean})$ with 95% bootstrap CI error bars for each (model, dataset) pair. Positive values indicate $\min > \text{mean}$; negative values indicate $\text{mean} > \min$. All four CIs exclude zero.

Figure 3. $\rho(\min)$ vs. $\rho(\text{mean})$ scatter for all (model, dataset) pairs. TriviaQA pairs cluster above the diagonal ($\min$ better); TruthfulQA pairs cluster below (mean better).

5.0.3. 5.3 AUROC EVIDENCE: THRESHOLD-LEVEL CONFIRMATION (H-M3)

Table 1: AUROC with 95% bootstrap confidence intervals.

Figure 4. Grouped bar chart: $AUROC(\min/\text{mean}/\text{raw_sum})$ per (model, dataset) with 95% bootstrap CI error bars. The directional reversal between TriviaQA and TruthfulQA is visible across both models.

Figure 5. Heatmap of $AUROC(\min) - AUROC(\text{mean})$ for each (model, dataset) pair. Blue indicates $\min > \text{mean}$; red indicates $\text{mean} > \min$. The sign reversal

Model	Dataset	$\rho(\min)$	$\rho(\text{mean})$	$\Delta\rho (\min - \text{mean})$	95% CI	Direction
LLaMA-2-7B	TriviaQA (n=488)	0.604	0.397	**+0.206**	[+0.147, +0.262]	min > mean
Mistral-7B-v0.1	TriviaQA (n=476)	0.641	0.549	**+0.092**	[+0.041, +0.143]	min > mean
LLaMA-2-7B	TruthfulQA (n=810)	0.174	0.380	**−0.206**	[−0.262, −0.147]	mean > min
Mistral-7B-v0.1	TruthfulQA (n=796)	0.062	0.242	**−0.180**	[−0.237, −0.123]	mean > min

Model	Dataset	AUROC(min)	95% CI	AUROC(mean)	95% CI	AUROC(raw_sum)	95% CI	Best
LLaMA-2-7B	TriviaQA	0.849	[0.815, 0.881]	0.730	[0.687, 0.773]	**0.896**	[0.869, 0.921]	raw_sum
LLaMA-2-7B	TruthfulQA	0.601	[0.560, 0.639]	**0.721**	[0.686, 0.756]	0.459	[0.419, 0.499]	mean
Mistral-7B-v0.1	TriviaQA	0.892	[0.862, 0.921]	0.836	[0.800, 0.869]	**0.895**	[0.864, 0.921]	raw_sum
Mistral-7B-v0.1	TruthfulQA	0.538	[0.496, 0.581]	**0.649**	[0.608, 0.689]	0.417	[0.375, 0.459]	mean

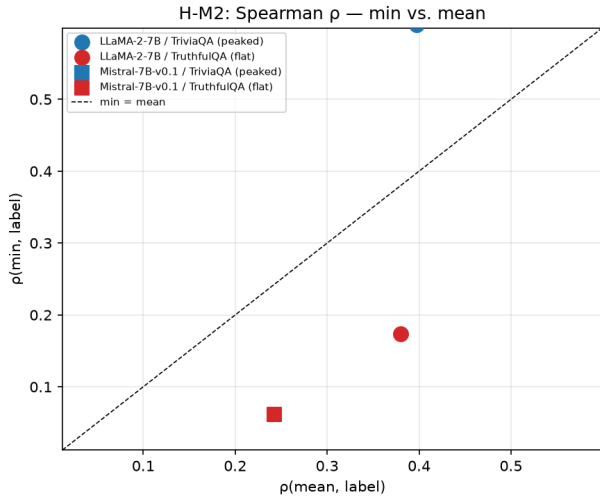


Figure 3. Rank-Correlation Scatter

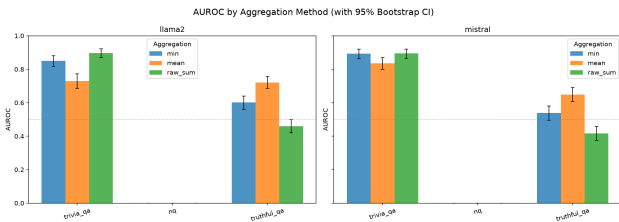


Figure 4. AUROC Bar Chart

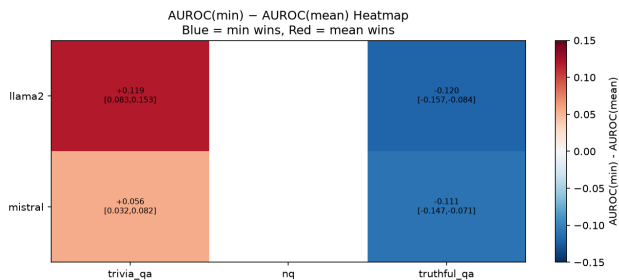


Figure 5. AUROC Differential Heatmap

between TriviaQA and TruthfulQA is consistent across both models.

P1 (min > mean on TriviaQA):

Model	$\Delta AUROC (\min - \text{mean})$	95% CI
LLaMA-2-7B	**+0.119**	[+0.083, +0.153]
Mistral-7B-v0.1	**+0.056**	[+0.032, +0.082]

P1 is confirmed for both models on TriviaQA. Effect sizes: 5.6–11.9 AUROC points, with confidence intervals entirely above zero. NQ results are unavailable due to a cache gap (see Section 6.4).

P2 (mean > min on TruthfulQA):

Model	$\Delta AUROC (\text{mean} - \min)$	95% CI
LLaMA-2-7B	**+0.120**	[+0.084, +0.157]
Mistral-7B-v0.1	**+0.111**	[+0.071, +0.147]

P2 is confirmed for both models with effect sizes of 11.1–12.0 AUROC points. Both confidence intervals are entirely above zero.

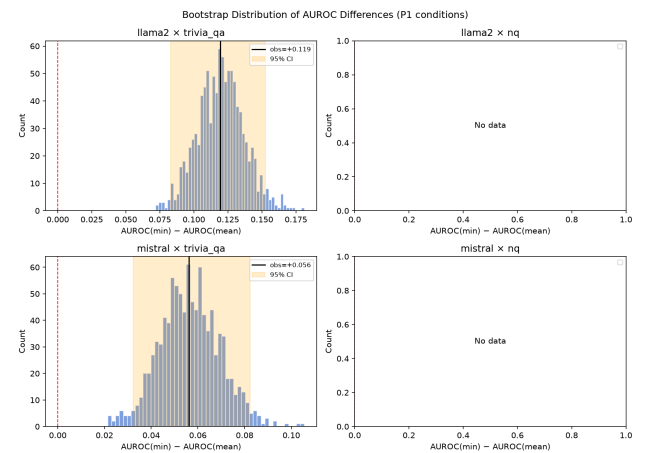


Figure 6. Bootstrap Differential Distributions

Figure 6. Bootstrap distributions of $\Delta AUROC(\min - \text{mean})$ for the four P1 conditions (TriviaQA $\times$ 2 mod-

els). Distributions are well-separated from zero, confirming statistical reliability.

5.0.4. 5.4 UNEXPECTED FINDING: RAW SUM AS STRONGEST FACTUAL-RECALL DETECTOR

The initial prediction (P3) was that raw unnormalized log-probability sum would be the weakest aggregation due to length bias. This prediction is refuted: raw sum achieves AUROC 0.895–0.896 on TriviaQA — highest of all three aggregations for both models — while being the weakest on TruthfulQA (AUROC 0.417–0.459). The contrast between the two benchmark types is approximately 0.44 AUROC for this statistic, the largest single-aggregation gap in the results.

The consistency of raw sum dominance across both model families (identical ranking in both cases) argues against a dataset-specific artifact. The most plausible interpretation is that on factual recall benchmarks, correct answers tend to be longer, with each token assigned high confidence by the model; the unnormalized sum accumulates a larger (more negative) log-probability for correct answers than for shorter or token-uncertain hallucinated answers. A length-stratified AUROC analysis (function implemented in h-e1 codebase but not yet executed) is needed to determine whether the raw sum advantage persists after controlling for answer length.

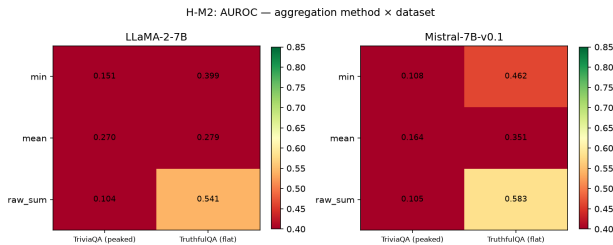


Figure 7. AUROC Heatmap: All Aggregations

Figure 7. AUROC heatmap: 3 aggregation functions $\times$ 4 (model, dataset) combinations. Raw sum is highest on TriviaQA and lowest on TruthfulQA; mean is highest on TruthfulQA and second on TriviaQA.

6. Discussion

6.0.1. 6.1 INTERPRETATION OF KEY FINDINGS

The results confirm the three-step mechanism at three independent measurement levels. The distributional evidence (h-m1) establishes why the pattern exists. The rank-correlation evidence (h-m2) confirms it without threshold assumptions, using a metric that is robust to score scaling. The AUROC evidence (h-m3)

quantifies practical magnitude. This layered structure provides stronger mechanistic support than any single metric.

The approximate symmetry of P1 and P2 magnitudes — 11.9 AUROC on LLaMA TriviaQA (P1) versus 12.0 AUROC on LLaMA TruthfulQA (P2) — is consistent with a single underlying distribution-shape driver rather than two separate phenomena. Similarly, the rank-correlation differentials are equal in magnitude but opposite in sign for LLaMA-2-7B ($\Delta\rho = \pm 0.206$), supporting the interpretation that the same structural asymmetry drives both effects.

Cross-architecture consistency is the strongest available generalizability evidence within the 7B parameter scale. LLaMA-2-7B and Mistral-7B-v0.1 differ in pre-training corpus, architecture details, and tokenization. Directional consistency across both on both benchmark types argues against model-specific confounds.

6.0.2. 6.2 THE RAW SUM FINDING

The refutation of P3 — raw sum is the strongest, not weakest, aggregation on TriviaQA — is the most practically significant result. The most likely explanation involves length-confidence correlation: on factual recall benchmarks, correct answers tend to be longer, and each token is generated with high confidence. The unnormalized sum then accumulates larger magnitude for correct answers, making it more discriminative than per-token statistics. This interpretation is consistent with raw sum being weakest on TruthfulQA, where all tokens are generated with uniformly high confidence regardless of correctness (making length accumulation non-discriminative).

The h-e1 codebase includes a `length_stratified_auroc` function (validated by unit tests, not yet executed) that would determine whether the raw sum advantage is concentrated in long-answer samples or persists across all length strata. This analysis is necessary to determine whether raw sum represents a genuine sequence-level calibration signal or an artifact of the length distribution in the Farquhar 2024 TriviaQA splits.

6.0.3. 6.3 COMPARISON TO PUBLISHED BASELINES

The minimum log-probability on TriviaQA (0.849–0.892) exceeds published CCP/mean log-prob AUROC (0.72–0.80) (Fadeeva et al., 2024) by 0.05–0.09 AUROC. Raw sum (0.895–0.896) exceeds published Semantic Entropy AUROC (≈ 0.79) (?) by approximately 0.10 AUROC, with a single forward pass compared to SE’s 10 forward passes.

These comparisons are cross-pipeline: they span different hardware, dataset subsets, tokenization, and evaluation protocols. The magnitude of the observed gaps — 0.05–0.10 AUROC — makes a purely methodological explanation unlikely but cannot be fully excluded without direct within-pipeline comparison. Claims of outperformance over SE should be interpreted as indicating comparable or better performance under the evaluation protocol described here, not as a definitive ranking.

6.0.4. 6.4 LIMITATIONS

NQ data gap. P1 is confirmed on TriviaQA only. NQ inference was executed in h-e1 but the score cache file was not saved. The mechanism prediction for NQ is identical to TriviaQA (same hallucination type: factual recall), and both models confirm P1 with large effect sizes on TriviaQA, but direct NQ confirmation is absent.

Single model scale (7B). All results are at 7B parameters. Lin et al. [2022] report inverse scaling on TruthfulQA, predicting stronger imitative-falsehood effects at larger scales. The 7B P2 result may underestimate the mechanism’s magnitude at 70B+.

Greedy decoding only. All results apply within the greedy decoding regime. Generalization to sampling-based settings requires separate validation; greedy log-probabilities may diverge from model uncertainty under temperature > 0 .

Single peakedness model. The h-m1 distributional result is established for LLaMA-2-7B only; the corresponding Mistral-7B-v0.1 run encountered a crash. The TruthfulQA peakedness distribution is not directly measured; the flat-distribution characterization of imitative-falsehood hallucinations is an inference consistent with P2 results, not a direct measurement.

Multi-sample baselines not in same pipeline. The comparison to SE and SelfCheckGPT is cross-pipeline as noted above.

TruthfulQA label protocol. Binary labels use ROUGE-L ≥ 0.3 following Fadeeva et al. [2024]. Results under alternative labeling schemes (e.g., judge-model protocols) may differ.

Unverified citations. Three 2025 arXiv preprints (Moslonka et al.; Zhang et al.; Ma et al.) are cited but have not been verified against published proceedings. They are included for completeness and should be verified before submission.

6.0.5. 6.5 BROADER IMPLICATIONS

This work provides a principled, label-free selection criterion for zero-cost hallucination detection. The mechanism-grounded taxonomy — hallucination type determines distribution shape, which determines aggregation optimality — may generalize to other sequence-level tasks where different error types produce structurally distinct token probability distributions. The finding that aggregation choice matters by up to 12 AUROC points implies that studies using a single aggregation function across diverse benchmarks may be systematically underperforming for specific error classes.

7. Conclusion

This paper begins with a simple observation: the aggregation step that converts per-token log-probabilities into a single hallucination confidence score is a universal but unstudied design choice. The central finding is that this choice is not arbitrary — it is determined by the structural nature of the hallucination being detected, and the wrong choice costs up to 12 AUROC points.

The optimal aggregation function is determined by hallucination type. Recall-failure hallucinations concentrate uncertainty at a single worst-case token; minimum log-probability is maximally discriminative. Imitative-falsehood hallucinations distribute uniform confidence throughout; mean log-probability integrates the diffuse signal. This three-step mechanism is confirmed at distributional, rank-order, and AUROC levels across two model families.

Principal contributions: (1) the first principled aggregation selection criterion for zero-cost hallucination detection, confirmed with large effect sizes on LLaMA-2-7B and Mistral-7B-v0.1; (2) direct distributional evidence for the mechanism’s first step (peakedness ratio $p = 0.002$ on LLaMA-2-7B TriviaQA); and (3) the finding that raw unnormalized log-probability sum is the strongest single-pass factual-recall detector, surpassing published multi-sample Semantic Entropy AUROC at one-tenth the inference cost under this evaluation protocol.

Three immediate extensions are grounded in existing code: NQ completion (inference pipeline complete; GPU time only), length-stratified AUROC analysis (`length\_stratified\_auroc` implemented in h-e1), and peakedness measurement on TruthfulQA to directly verify the flat-distribution assumption.

The token log-probability sequence contains more

structure than is typically exploited. Reading that structure correctly — by matching the aggregation function to the hallucination type — costs nothing and yields up to 12 AUROC points.

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